Business Information Systems

Week 05 — EIS (Enterprise Information System) Acquisition

Main Decisions · Roles & Responsibilities · Business Case · Risk Management · Ethical Aspects

From the Industry to the Classroom — Building the IT's Future, Today and Together!

Today's roadmap: the five links of the chain

1. **Main decisions**
   Build it, buy it, or rent it? The forks that shape everything.

2. **Roles & responsibilities**
   Who decides, who does the work, who signs off.

3. **Business case**
   Does the money actually make sense?

4. **Risk management**
   What could go wrong — and our plan for it.

5. **Ethical aspects**
   Should we do this, even if we can?

OUR RUNNING EXAMPLE

Meet Northern Roast Coffee Co.

– A small Montréal coffee roaster that grew fast — from a garage to four cafés.

– Eleven staff, two wholesale clients, and a notebook that became forty spreadsheets.

– None of them agree. Inventory says 40 kg of beans; the café ran out yesterday.

– Maya, the owner, doesn't need another spreadsheet. She needs one connected system: an EIS.

– The question isn't whether she needs one — it's how to get it without breaking the company.

4 cafés

11 staff

40 spreadsheets

THE BIG IDEA

What is an EIS?

An EIS (Enterprise Information System) is one large, shared platform that runs the core of a business and keeps a single, trusted version of the data — instead of many disconnected apps.

ERP

Enterprise Resource Planning

Finance, inventory, purchasing, operations. e.g. SAP, Oracle, MS Dynamics.

CRM

Customer Relationship Management

Customers, leads, support. e.g. Salesforce.

SCM

Supply Chain Management

Suppliers, warehouses, logistics — the flow of goods.

Internal vs external integration

Internal integration

- Connecting your OWN departments to each other.
- Sales talks to inventory; inventory talks to finance.
- Turns 40 quarrelling spreadsheets into one source of truth.
- Nobody asks 'whose number is right?'

External integration

- Connecting your system to the OUTSIDE world.
- Suppliers, banks, payment processors, tax filings.
- Usually rides on an API (Application Programming Interface).
- How you ACQUIRE the system decides how easy this is later.

Lesson 1: EIS Acquisition

From 'we think we need a system' to 'it's live and people use it' — in five links.

1

Main decisions

The big, hard-to-reverse choices that shape everything after.

The headline decision: build, buy, or rent?

Build

Write custom software from scratch. Perfect fit, you own it forever — and 'forever' is how long you maintain it.

Buy — COTS

License finished Commercial Off-The-Shelf software, install on your own servers, configure to fit.

Rent — SaaS

Software as a Service over the internet for a monthly fee. The vendor runs servers, backups and updates.

Two more forks

CapEx (up-front asset) vs OpEx (ongoing subscription); one suite vs best-of-breed tools you stitch together.

How professionals choose (not on vibes)

- **Weighted decision matrix:** list the criteria, weight each by importance, score every option 1–5, let the arithmetic pick the winner.
- **TCO (Total Cost of Ownership):** count the WHOLE multi-year cost — licences, training, support, upgrades — not just the sticker price.

RFI

Request for Information

'Tell us what you've got.'

RFP

Request for Proposal

'Here are our needs — propose a solution & price.'

RFQ

Request for Quotation

'We know what we want — just quote it.'

Example: Northern Roast picks its path

What this example does: Runs Maya's three options (Build, Buy, SaaS) through a weighted decision matrix to produce a defensible recommendation.

How it relates to the lecture: Demonstrates the MAIN DECISION of acquisition — choosing a path with evidence, not instinct.

- Eight weighted criteria: fit, 5-yr TCO, speed, scalability, vendor lock-in, security & PIPEDA fit, demand on the IT team, maintenance.
- Each option scored 1–5 on every criterion; the weighted totals decide the winner.

The build / buy / SaaS decision tree

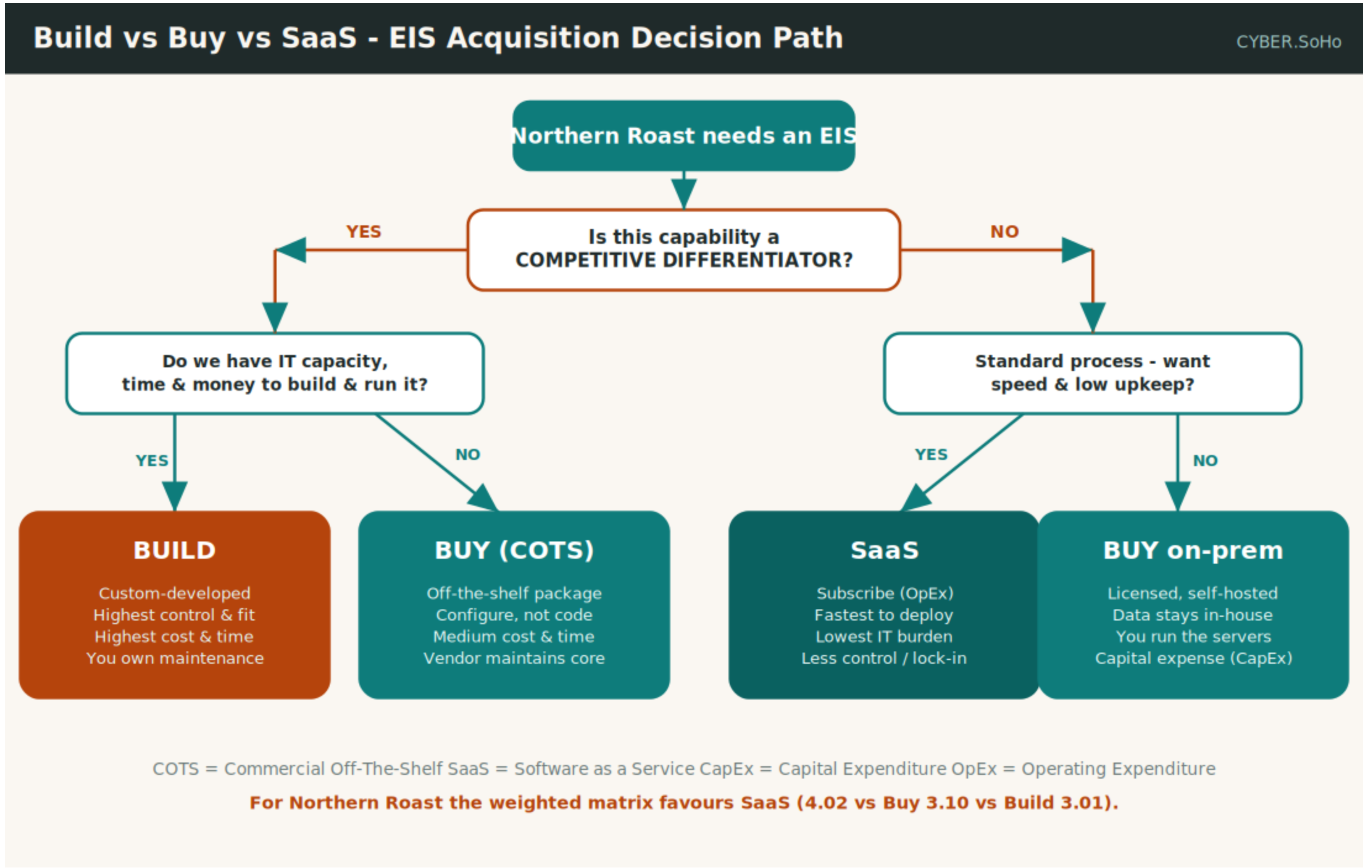

Figure 1 — Northern Roast's acquisition decision tree (illustrative). Source: Cyber.SoHo.

The scored decision matrix (weighted)

Decision Criterion	Weight	Build	Buy	SaaS
Fit to processes	0.20	1.00	0.60	0.60
5-yr Total Cost of Ownership	0.18	0.36	0.54	0.72
Speed to deploy	0.15	0.15	0.45	0.75
Scalability as we grow	0.12	0.48	0.36	0.60
Vendor lock-in / exit ease	0.10	0.50	0.30	0.20
Security & compliance (PIPEDA)	0.10	0.30	0.40	0.40
Demand on small IT team	0.08	0.08	0.24	0.40
Ongoing maintenance burden	0.07	0.14	0.21	0.35
TOTAL (weighted)	**1.00**	**3.01**	**3.10**	**4.02**

Weighted score = score × weight. Preview of 01_weighted_decision_matrix.xlsx.

And the winner is…

3.01	**3.10**	**4.02**
Build	Buy (COTS)	SaaS

SaaS wins — fast to deploy, scales easily, barely touches the IT team. The matrix made the trade-offs visible.

Watch (≤ 15 min): what an ERP system actually is

"What is ERP?" — a short, stable explainer of the most common kind of EIS

https://www.youtube.com/watch?v=PVRgIXLWDHs

Build vs buy (search): youtube.com/results?search_query=build+vs+buy+software+enterprise+system

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2

Roles & responsibilities

More EIS projects die from confusion about who-does-what than from any technical flaw.

Who's in the room?

Sponsor (CFO)
Wants it, controls the money.

Steering Committee
Sets direction, unblocks obstacles.

Project Manager (PM)
Runs schedule, budget, cat-herding.

Business Analyst (BA)
Turns needs into precise requirements.

IT Lead
Owns data, security, integration.

Vendor
Sells and configures the system.

End Users
Live in it daily — asked last, matter most.

Change Manager
Training, comms, and the human side.

The tool: a RACI matrix

RACI = Responsible, Accountable, Consulted, Informed. One letter per person per task.

R — Responsible

Does the work. There can be several.

A — Accountable

Owns the outcome and signs off. Exactly ONE per task.

C — Consulted

Gives two-way input BEFORE the work happens.

I — Informed

Kept in the loop, one-way, after the fact.

The golden rule

Exactly ONE Accountable, at least ONE Responsible — on every single row.

Two A's?

Two people think they're in charge — and quietly contradict each other.

Zero R's?

Everyone assumes someone else is doing the work — so nobody is.

Example: Northern Roast assigns the work

What this example does: Assigns clear ownership for the ten key project tasks using a RACI matrix.

How it relates to the lecture: Brings ROLES & RESPONSIBILITIES to life — no decision falls through a crack.

- Maya is Accountable for money & direction (budget, vendor selection, go-live).
- The PM is Accountable for delivery (migration, testing, training). Every row validates: one A, at least one R.

Governance & RACI structure

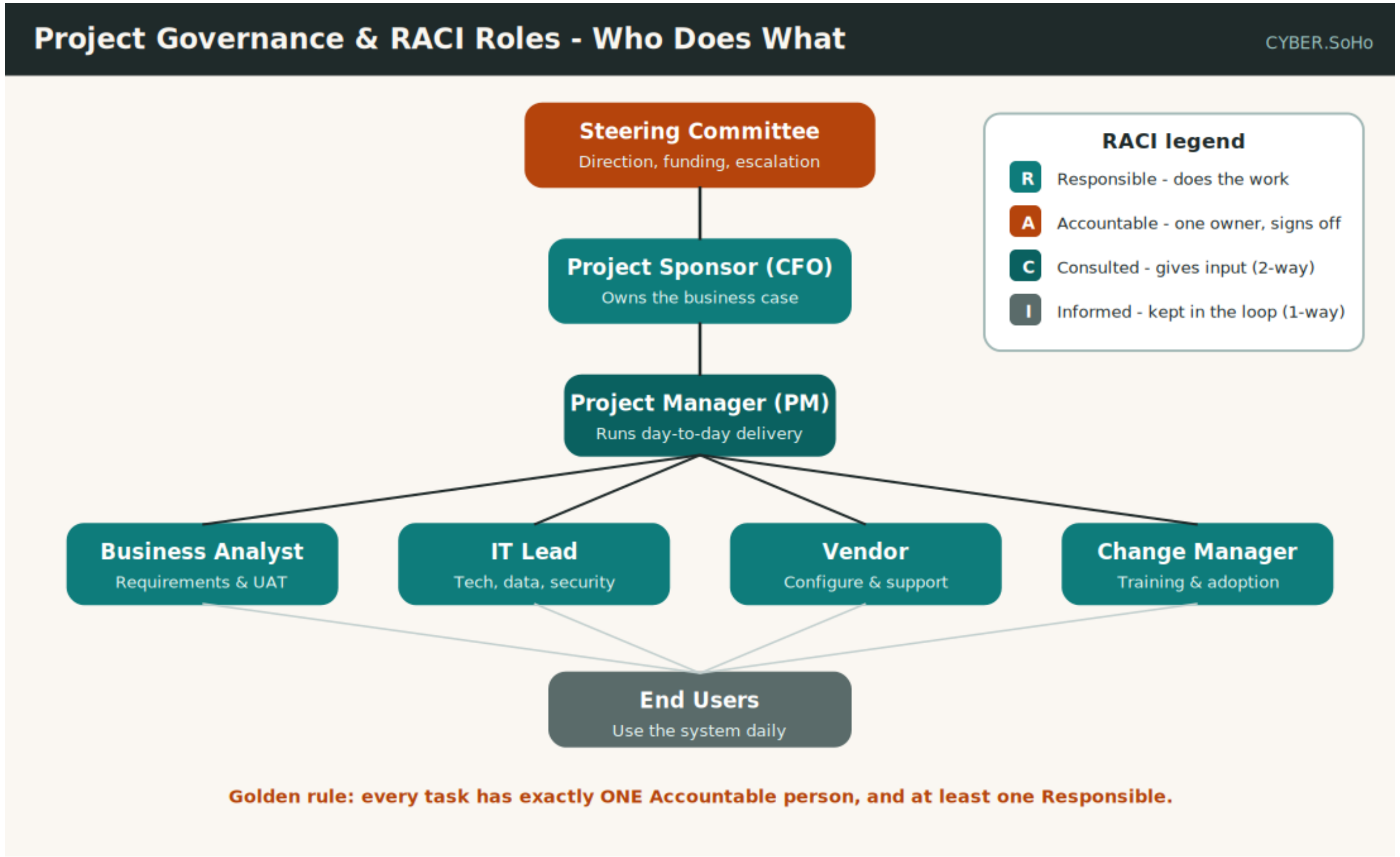

Figure 2 — Project governance and the RACI legend (illustrative). Source: Cyber.SoHo.

Northern Roast's RACI matrix

Task / Decision	Spon	Steer	PM	BA	IT	Vend	User	Chg
Define business requirements	A	C	R	R	C	I	C	C
Approve project budget	A	R	C	I	I		I	I
Shortlist & select vendor	A	C	R	C	C	I	I	I
Negotiate contract & SLA	A	C	R	I	C	C		I
Migrate & validate data	I	I	A	C	R	R	C	I
Configure / customise system	I	I	A	C	R	R	I	I
User Acceptance Testing (UAT)	I	I	A	R	C	C	R	C
Train the end users	I	I	A	C	I	C	R	R
Go-live go / no-go decision	A	C	R	C	C	C	I	C
Post-go-live support	I	I	A	I	R	C	C	R

R=Responsible · A=Accountable (one) · C=Consulted · I=Informed. All ten rows validate. Preview of 02_raci_matrix.xlsx.

Watch (≤ 15 min): the RACI matrix explained

"What is a RACI Matrix?" — a clear beginner walkthrough

https://www.youtube.com/watch?v=TMT_WPFh6RU

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3

Business case

Does the money make sense? No case, no cheque.

Costs vs benefits

Costs (easy to undercount)

– Up-front: licences, implementation, data migration, training.

– Annual long tail: subscription, support, upgrades.

– Always use TCO (Total Cost of Ownership), never the sticker price.

Benefits (two kinds)

– Tangible: hours saved, fewer stock errors, faster reporting — put a dollar on it.

– Intangible: better decisions, happier staff, a professional brand.

– Quantify the tangible honestly; name the intangible without faking a price.

Four money measures to know cold

ROI — Return on Investment

Net gain ÷ cost, as a %. 'For every dollar spent, how many come back?'

NPV — Net Present Value

Future cash discounted to today's value, summed. Positive NPV creates value.

IRR — Internal Rate of Return

The discount rate at which NPV = 0. Higher is better.

Payback period

How long until you get your money back. Discounted payback is the honest version.

Example: Northern Roast builds the business case

What this example does: Turns Maya's hunch into a rigorous five-year model computing TCO, NPV, ROI and payback.

How it relates to the lecture: The BUSINESS CASE made concrete — the artefact that unlocks (or blocks) the budget.

– Costs: $120,000 to implement (Year 0) + $28,000/yr to run.

– Benefits: 2,600 hours saved × $32 = $83,200, plus $41,000 fewer stock errors + $18,000 faster reporting = $142,200/yr. Discounted at 10%.

The business-case value timeline

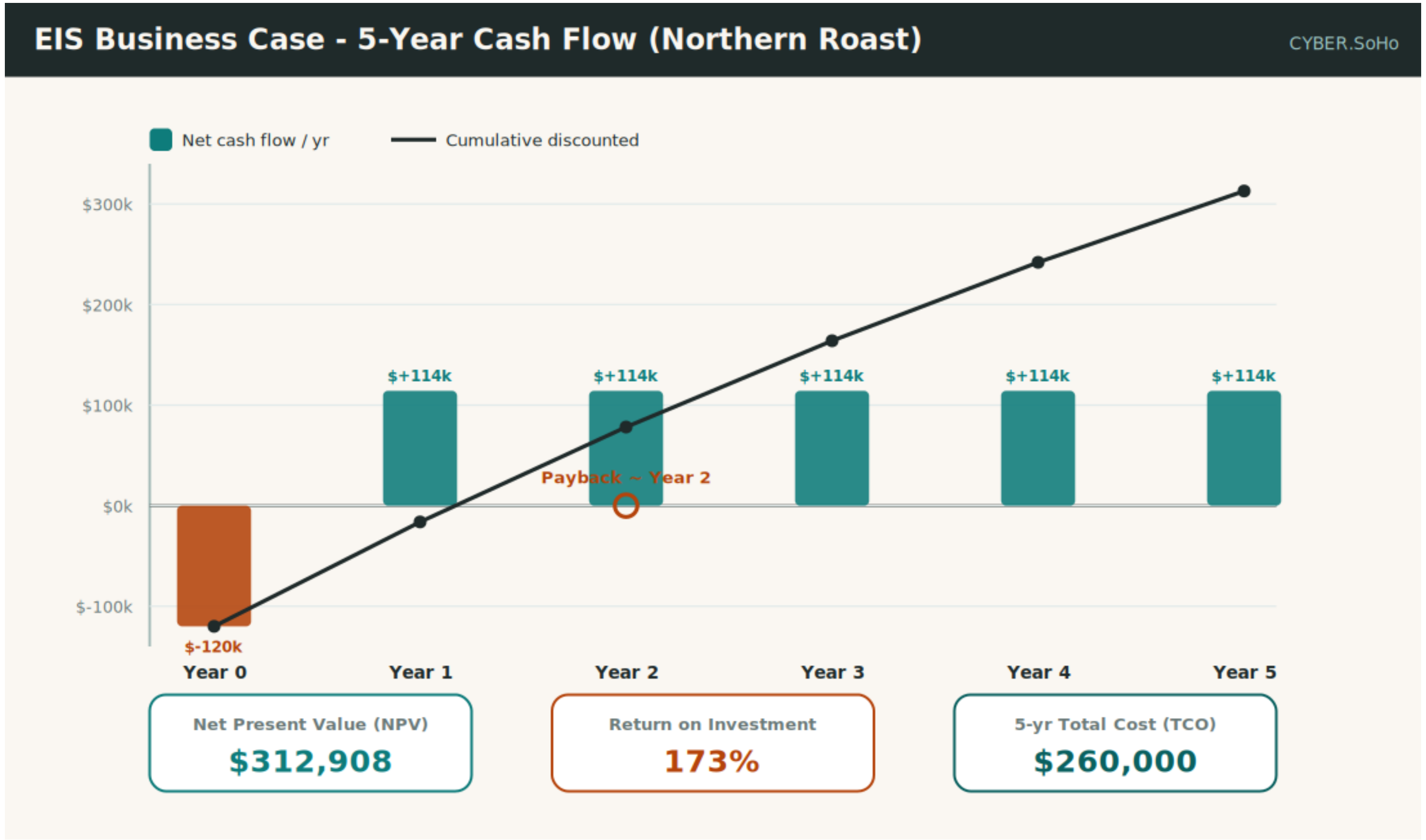

Figure 3 — Net cash flow, cumulative value, and the payback point (illustrative). Source: Cyber.SoHo.

The five-year cash-flow model

Line item ($)	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5
Costs (cash out)	-120,000	-28,000	-28,000	-28,000	-28,000	-28,000
Benefits (cash in)	0	142,200	142,200	142,200	142,200	142,200
Net cash flow	**-120,000**	**114,200**	**114,200**	**114,200**	**114,200**	**114,200**
Cumulative discounted	-120,000	-16,182	78,198	163,998	241,999	312,908

Discounted at 10%. Cumulative turns positive in Year 2. Preview of 03_business_case_tco_npv.xlsx.

The verdict: a clear GO

$312,908

Net Present Value

173%

Return on Investment

Year 2

Discounted payback

Total Cost of Ownership $260,000 vs $711,000 of benefits over five years. A sponsor says yes to this.

Watch (≤ 15 min): NPV, ROI & payback for beginners

Curated search — pick a short explainer from Corporate Finance Institute or 365 Financial Analyst

youtube.com/results?search_query=NPV+ROI+payback+period+explained+for+beginners

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4

Risk management

Optimism gets the project approved. Risk management gets it delivered.

What is a risk?

Risk score = Probability × Impact (each scored 1–5)

- A risk is a future event that MIGHT happen and WOULD hurt if it did.
- Multiplying stops you over-worrying dramatic-but-unlikely disasters and under-worrying boring-but-frequent ones.
- **Risk register:** a living table — each risk, its score, severity band, owner, and planned action.
- **Heat map:** a 5×5 grid (probability × impact) shaded green to red. One glance shows what to tackle first.

Four ways to respond (the 'four T's')

Reduce (Treat)

Lower the probability or the impact with a mitigation.

Transfer

Hand it to someone else — insurance, or push it into the vendor's contract.

Accept (Tolerate)

Too small or too costly to fix — accept it on purpose, with eyes open.

Avoid (Terminate)

Change the plan so the risk simply cannot occur.

Example: Northern Roast's risk register

What this example does: Identifies ten realistic risks, scores each by probability × impact, assigns an owner and a mitigation.

How it relates to the lecture: RISK MANAGEMENT in action — vague anxiety becomes a prioritised, owned, actionable list.

- Three risks land in the High band; one is small enough to simply accept.
- Notice that two of the three High risks (lock-in, low adoption) are PEOPLE problems, not technology ones.

The 5×5 risk heat map

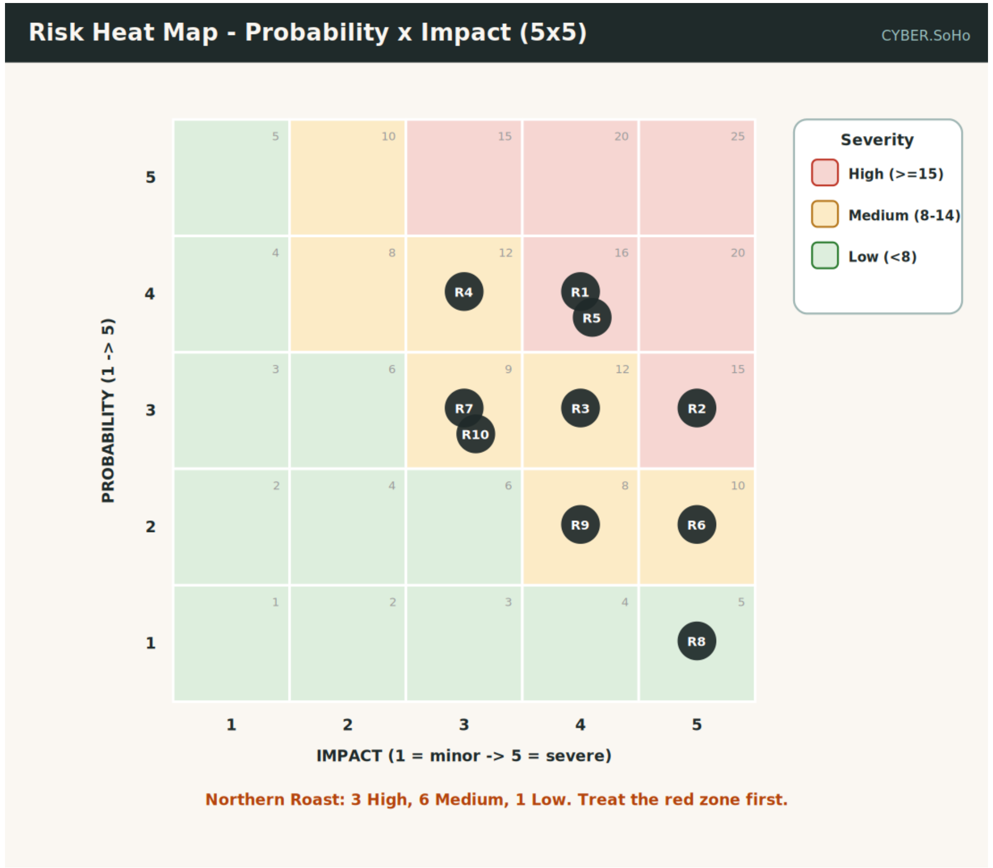

Figure 4 — Probability × impact heat map with the ten risks plotted (illustrative). Source: Cyber.SoHo.

The risk register

ID	Risk	Prob	Impact	Score	Severity	Response
R1	Vendor lock-in (hard to leave)	4	4	16	**High**	Reduce
R2	Data migration corrupts records	3	5	15	**High**	Reduce
R3	Budget overrun > 20%	3	4	12	Medium	Reduce
R4	Scope creep delays go-live	4	3	12	Medium	Reduce
R5	Low user adoption	4	4	16	**High**	Reduce
R6	Security / privacy breach	2	5	10	Medium	Transfer
R7	Integration with POS fails	3	3	9	Medium	Reduce
R8	Vendor goes insolvent	1	5	5	Low	Accept
R9	PIPEDA non-compliance	2	4	8	Medium	Reduce
R10	Key person leaves mid-project	3	3	9	Medium	Reduce

Score = Probability × Impact. ≥15 High · 8–14 Medium · <8 Low. Preview of 04_risk_register.xlsx.

Where the ten risks landed

3	6	1
High	Medium	Low

Lock-in, a botched migration, and staff not adopting it are the three to lose sleep over.

Watch (≤ 15 min): evaluating risk impact & probability

"How to Evaluate Risk Impact and Probability?" — a practical beginner guide

https://www.youtube.com/watch?v=XK1w42PXbWY

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5

Ethical aspects

Risk asks 'what could go wrong for us?' Ethics asks 'for everyone else?'

Should we — not just can we?

An EIS holds people's data and increasingly makes automated decisions about them. That is power — and power without ethics is just a faster way to do harm. Start with privacy.

Privacy: the law and two principles

- **PIPEDA (Personal Information Protection and Electronic Documents Act):** Canada's privacy law, enforced by the Office of the Privacy Commissioner.
- **GDPR (General Data Protection Regulation):** applies if you hold data on people in Europe.
- **Data minimisation:** collect only what you need (does a coffee app really need a birth date?).
- **Purpose limitation + informed consent:** use data only for the stated reason; ask in plain language.

Four more dimensions of doing it right

Fairness & bias

Explainability (can it say WHY it decided?) and bias testing of automated decisions.

Transparency

A documented, defensible procurement — free of kickbacks and quiet conflicts.

Accessibility (WCAG)

Works for everyone, including staff with disabilities — the W3C's guidelines.

Sustainability

Energy and carbon of the data centres, and labour conditions in the tech supply chain.

Example: Northern Roast runs an ethics gate

What this example does: Scores the choice against ten weighted ethical questions, then lands in a clear go / no-go band.

How it relates to the lecture: ETHICAL ASPECTS made operational — a concrete gate Maya must clear before signing.

- Each question answered Yes (2), Partial (1), or No (0), weighted by seriousness.
- Ethics is a GATE the project passes through before signing — not a sticker added afterward.

The ethics decision framework

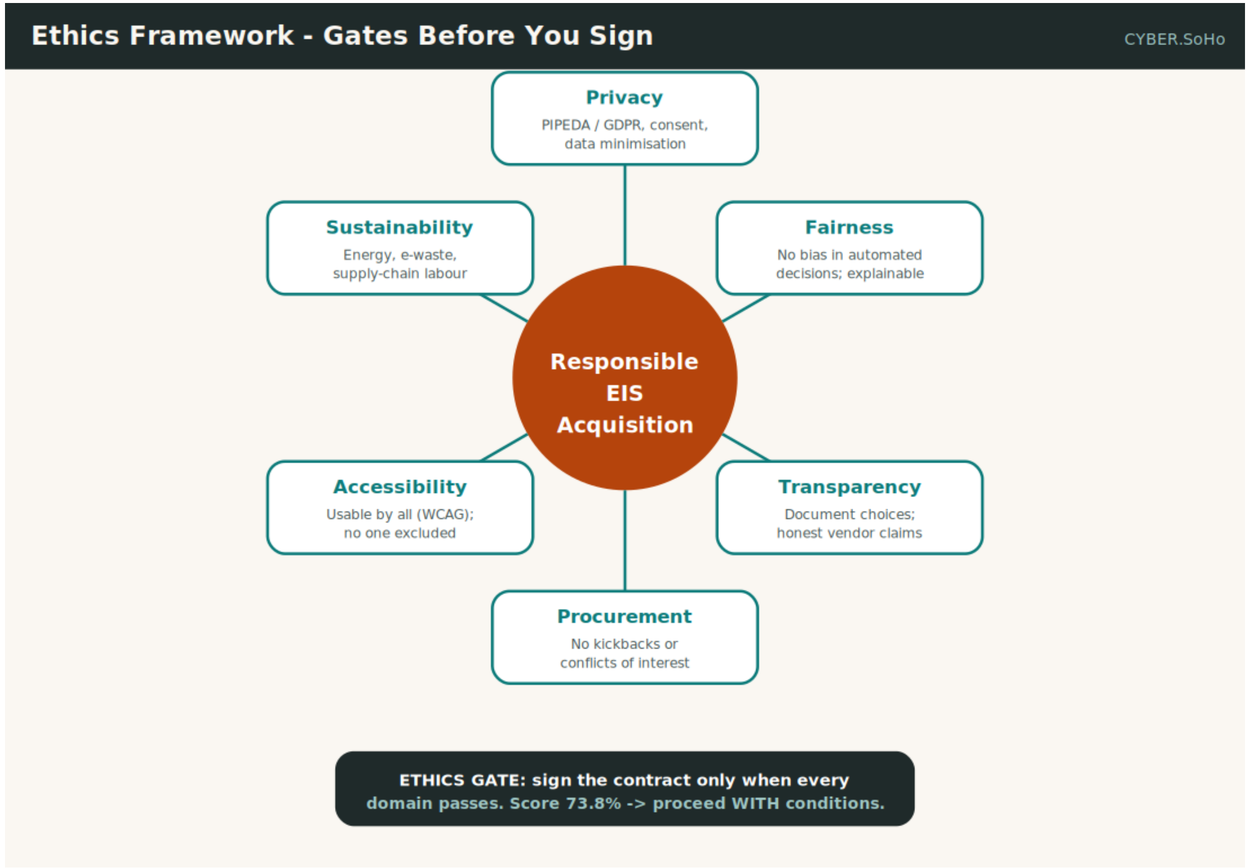

Figure 5 — Six ethical domains feeding one responsible-acquisition gate (illustrative). Source: Cyber.SoHo.

The ethics result

31 / 42	73.8%	Proceed
points scored	ethics readiness	with conditions

Not unethical, not ready to sign. Fix the weak spots — consent, bias testing, accessibility, data residency — first.

Watch (≤ 15 min): ethical data — privacy, bias & responsible use

"Ethical Big Data: Privacy, Bias, & Responsible Use for Beginners"

https://www.youtube.com/watch?v=BQr29XIsvF0

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Wrapping up — the chain

Not a purchase, but a discipline. Acquire well, and the doors stay open.

What Maya actually did

1. **Main decision:** Chose SaaS with evidence — weighted score 4.02.
2. **Roles:** Gave every task an owner with a RACI matrix — nothing fell through a crack.
3. **Business case:** Proved a positive NPV of ~$312,908 and a Year 2 payback.
4. **Risk:** Flagged 3 High risks before they could bite.
5. **Ethics:** Refused to sign until the 73.8% gate's weak spots were fixed.

The technology was rarely the hard part — the decisions, the people, and the judgement were.

Reference library

Vendors & bodies

- SAP, Oracle, Microsoft Dynamics 365 — ERP suites.
- Salesforce — the best-known CRM (SaaS).
- Project Management Institute (PMI) — governance.
- ISACA, NIST, ISO — risk & security frameworks.
- Gartner — vendor research & comparisons.
- Corporate Finance Institute — ROI / NPV / IRR.

Privacy, ethics & safety

- Office of the Privacy Commissioner of Canada — PIPEDA.
- European Commission — GDPR.
- W3C Web Accessibility Initiative — WCAG.
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